1.46 Proof by Induction: Matrices

Worked Solutions

1 a)

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} = \begin{pmatrix} 6k+3-4k & -12k-4+8k \\ 2k+1-k & -4k-1+2k \end{pmatrix}$$

$$= \begin{pmatrix} 2k+3 & -4k-4 \\ k+1 & -2k-1 \end{pmatrix} = \begin{pmatrix} 2(k+1)+1 & -4(k+1) \\ k+1 & -2(k+1)+1 \end{pmatrix} \text{ as required.}$$

1 b)

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2^k-1 \\ 0 & 2^k \end{pmatrix} = \begin{pmatrix} 1 & 2^k-1+2^k \\ 0 & 2\times 2^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2^{k+1}-1 \\ 0 & 2^{k+1} \end{pmatrix} \text{ as required.}$$

1 c)

$$\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^k-1) & 5^k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1-\frac{5}{4}(5^k-1) & 5\times 5^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -1-\frac{5}{4}\times 5^k+\frac{5}{4} & 5^{k+1} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}+\frac{1}{4}5^k & 5^{k+1} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1} \end{pmatrix} \text{ as required.}$$

2)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

3)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 2 \times 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 + 2k \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2(k+1) \\ 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

4)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 2+1 & -2 \\ 2 & 1-2 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 2k+1 & -2k \\ 2k & 1-2k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2k+1 & -2k \\ 2k & 1-2k \end{pmatrix}$$

$$= \begin{pmatrix} 6k+3-4k & -6k-2+4k \\ 4k+2-2k & -4k-1+2k \end{pmatrix} = \begin{pmatrix} 2k-2 & -2k-2 \\ 2k+2 & -2k-1 \end{pmatrix}$$

$$= \begin{pmatrix} 2(k+1)+1 & -2(k+1) \\ 2(k+1) & 1-2(k+1) \end{pmatrix}$$

which is $\begin{pmatrix} 2n+1 & -2n \\ 2n & 1-2n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

5)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 2+1 & 2i \\ 2i & 1-2 \end{pmatrix} = \begin{pmatrix} 3 & 2i \\ 2i & -1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 2k+1 & 2ki \\ 2ki & 1-2k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 3 & 2i \\ 2i & -1 \end{pmatrix} \begin{pmatrix} 2k+1 & 2ki \\ 2ki & 1-2k \end{pmatrix}$$

$$= \begin{pmatrix} 6k+3-4k & 6ki+2i-4ki \\ 4ki+2i-2ki & -4k-1+2k \end{pmatrix} = \begin{pmatrix} 2k+3 & 2ki+2i \\ 2ki+2i & -1-2k \end{pmatrix}$$

$$= \begin{pmatrix} 2(k+1)+1 & 2(k+1)i \\ 2(k+1)i & 1-2(k+1) \end{pmatrix}$$

which is $\begin{pmatrix} 2n+1 & 2ni \\ 2ni & 1-2n \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

6)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 2^1 & 4^1 - 2^1 \\ 0 & 4^1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 2^k & 4^k - 2^k \\ 0 & 4^k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 2^k & 4^k - 2^k \\ 0 & 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 2 \times 2^k & 2 \times 4^k - 2 \times 2^k + 2 \times 4^k \\ 0 & 4 \times 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 2^1 \times 2^k & 4^1 \times 4^k - 2^1 \times 2^k \\ 0 & 4^1 \times 4^k \end{pmatrix} = \begin{pmatrix} 2^{k+1} & 4^{k+1} - 2^{k+1} \\ 0 & 4^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 2^n & 4^n - 2^n \\ 0 & 4^n \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

7)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 2^1 & 1 \times 2^{1-1} \\ 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 2^k & k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2^k & k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}$$

$$= \begin{pmatrix} 2 \times 2^k & 2k \times 2^{k-1} + 2^k \\ 0 & 2 \times 2^k \end{pmatrix} = \begin{pmatrix} 2^1 \times 2^k & k \times 2^k + 2^k \\ 0 & 2^1 \times 2^k \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & (k+1) \times 2^k \\ 0 & 2^{k+1} \end{pmatrix} = \begin{pmatrix} 2^{k+1} & (k+1) \times 2^{(k+1)-1} \\ 0 & 2^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

8)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & \frac{1}{2}(3^1 - 1) \\ 0 & 3^1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) \\ 0 & 3^k \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) \\ 0 & 3^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1) + 3^k \\ 0 & 3 \times 3^k \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2}(3^k - 1 + 2 \times 3^k) \\ 0 & 3^1 \times 3^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \frac{1}{2}(3 \times 3^k - 1) \\ 0 & 3^{k+1} \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2}(3^{k+1} - 1) \\ 0 & 3^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) \\ 0 & 3^n \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1 - 4^1) & 4^1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^k) & 4^k \end{pmatrix}$$

Prove true for $n = k+1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^k) & 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -1+\frac{4}{3}(1-4^k) & 4 \times 4^k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1+\frac{4}{3}-\frac{1}{3}\times 4 \times 4^k & 4^1 \times 4^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ \frac{1}{3}-\frac{1}{3}\times 4^1 \times 4^k & 4^{k+1} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^{k+1}) & 4^{k+1} \end{pmatrix}$$

which is $\begin{pmatrix} 1 & 0 \\ \frac{1}{3}(1-4^n) & 4^n \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$.

10 a)

$$A^2 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 0 & 1 \end{pmatrix}$$

$$A^4 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 9 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 12 \\ 0 & 1 \end{pmatrix}.$$

10 b)

$$A^n = \begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}.$$

10 c)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 3 \times 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & 3k \\ 0 & 1 \end{pmatrix}$$

Prove true for $n = k+1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3k \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3k+3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3(k+1) \\ 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

11)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 4^1 & \frac{a}{3}(4^1 - 1) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 4^k & \frac{a}{3}(4^k - 1) \\ 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4^k & \frac{a}{3}(4^k - 1) \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \times 4^k & \frac{4}{3}a(4^k - 1) + a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4^1 \times 4^k & \frac{4}{3}a \times 4^k - \frac{4}{3}a + a \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4^{k+1} & \frac{a}{3} \times 4 \times 4^k - \frac{a}{3} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4^{k+1} & \frac{a}{3}(4^{k+1} - 1) \\ 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 4^n & \frac{a}{3}(4^n - 1) \\ 0 & 1 \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

12 a)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 - 3 & -9 \\ 1 & 3 + 1 \end{pmatrix} = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 - 3k & -9k \\ k & 3k + 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 - 3k & -9k \\ k & 3k + 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 + 6k - 9k & 18k - 27k - 9 \\ 1 - 3k + 4k & -9k + 12k + 4 \end{pmatrix} = \begin{pmatrix} -2 - 3k & -9k - 9 \\ 1 + k & 3k + 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 3(k+1) & -9(k+1) \\ k + 1 & 3(k+1) + 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 - 3n & -9n \\ n & 3n + 1 \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

12 b)

$$\begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$-2x - 9y = x \Rightarrow -9y = 3x \Rightarrow y = -\frac{1}{3}x$$

$$x + 4y = y \Rightarrow 3y = -x \Rightarrow y = -\frac{1}{3}x.$$

So any point on the line $y = -\frac{1}{3}x$ is invariant.

12 c)

$$\begin{pmatrix} 1-3n & -9n \\ n & 3n+1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$(1-3n)x - 9ny = x \Rightarrow -9ny = 3nx \Rightarrow y = -\frac{1}{3}x$$

$$nx + 3ny + y = y \Rightarrow 3ny = -nx \Rightarrow y = -\frac{1}{3}x.$$

So any point on the line $y = -\frac{1}{3}x$ is invariant.

13 a)

$$\det A = 4 + a^2 = 0$$

So $a = \pm 2i$.

13 b)

$$A = \begin{pmatrix} 2 & -2i \\ 2i & 2 \end{pmatrix} \quad \text{or} \quad A = \begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix}.$$

Prove true for $n = 1$:

$$A^1 = 4^{1-1}A = 4^0 A = A$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = 4^{k-1}A.$$

Prove true for $n = k + 1$:

If $a = -2i$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 2 & -2i \\ 2i & 2 \end{pmatrix} \times 4^{k-1} \begin{pmatrix} 2 & -2i \\ 2i & 2 \end{pmatrix}$$

$$= 4^{k-1} \begin{pmatrix} 4+4 & -4i-4i \\ 4i+4i & 4+4 \end{pmatrix} = 4^{k-1} \begin{pmatrix} 8 & -8i \\ 8i & 8 \end{pmatrix}$$

$$= 4 \times 4^{k-1} \begin{pmatrix} 2 & -2i \\ 2i & 2 \end{pmatrix} = 4^1 \times 4^{k-1} A = 4^k A$$

which is $4^{n-1}A$ with $k + 1$ substituted in.

If $a = 2i$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix} \times 4^{k-1} \begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix}$$
$$= 4^{k-1} \begin{pmatrix} 4+4 & 4i+4i \\ -4i-4i & 4+4 \end{pmatrix} = 4^{k-1} \begin{pmatrix} 8 & 8i \\ -8i & 8 \end{pmatrix}$$
$$= 4 \times 4^{k-1} \begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix} = 4^1 \times 4^{k-1} A = 4^k A$$

which is $4^{n-1}A$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

14)

Prove true for $n = 1$:

$$B^1AB^1 = B(AB) = B(B^4A) = B^5A = IA = A$$

so true for $n = 1$.

Assume true for $n = k$:

$$B^kAB^k = A.$$

Prove true for $n = k + 1$:

$$B^{k+1}AB^{k+1} = BB^kAB^kB$$
$$= B(B^kAB^k)B$$
$$= BAB$$
$$= B(B^4A)$$
$$= B^5A$$
$$= IA$$
$$= A.$$

So true for $n = k + 1$.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

15 a)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 1 & \frac{1}{2}(1+1) \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & k & \frac{1}{2}k(k+1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & \frac{1}{2}k(k+1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}k(k+1)+k+1 \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k(k+1)+2k+2) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k+1)(k+2) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

15 b)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 1 & 1 & \frac{1}{2}(1+3) \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 1 & k & \frac{1}{2}k(k+3) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & \frac{1}{2}k(k+3) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}k(k+3)+k+2 \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k(k+3)+2k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k^2+3k+2k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k^2+5k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k+1)(k+4) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 1 & n & \frac{1}{2}n(n+3) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

15 c)

Prove true for $n = 1$:

$$A^1 = \begin{pmatrix} 2^1 & 1-2^1 & 3(2^1-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so true for $n = 1$.

Assume true for $n = k$:

$$A^k = \begin{pmatrix} 2^k & 1-2^k & 3(2^k-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Prove true for $n = k + 1$:

$$A^{k+1} = A^1 \times A^k = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2^k & 1-2^k & 3(2^k-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \times 2^k & 2(1-2^k)-1 & 6(2^k-1)+3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^1 \times 2^k & 2-2\times 2^k-1 & 3(2\times 2^k-2)+3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 1-2^1\times 2^k & 3(2^1\times 2^k-2+1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 1-2^{k+1} & 3(2^{k+1}-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

which is $\begin{pmatrix} 2^n & 1-2^n & 3(2^n-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.